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Injectable and tissue adhesive EGCG-laden hyaluronic acid hydrogel depot for treating oxidative stress and inflammation

Zhichao He [a,b,c,1], Huitong Luo [a,b,1], Zetao Wang [a,b,d], Dafu Chen [e], Qi Feng [a,b,c,*], Xiaodong Cao [a,b,d,*]

[a] Department of Biomedical Engineering, School of Materials Science and Engineering, South China University of Technology, Guangzhou 510006, PR China
[b] National Engineering Research Center for Tissue Restoration and Reconstruction (NERC-TRR), South China University of Technology, Guangzhou 510006, PR China
[c] Key Laboratory of Biomedical Materials and Engineering of the Ministry of Education, South China University of Technology, Guangzhou 510006, PR China
[d] Key Laboratory of Biomedical Engineering of Guangdong Province, South China University of Technology, Guangzhou 510641, PR China
[e] Laboratory of Bone Tissue Engineering, Beijing Laboratory of Biomedical Materials, Beijing Research Institute of Orthopaedics and Traumatology, Beijing JiShuiTan Hospital, Beijing 100035, PR China

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ABSTRACT

Oxidative stress and inflammation are common pathological mechanisms for the progression of tissue degeneration. Epigallocatechin-3-gallate (EGCG) features antioxidant and anti-inflammatory properties, which is a promising drug for the treatment of tissue degeneration. Herein, we utilize the phenylborate ester reaction of EGCG and phenylboronic acid (PBA) to fabricate an injectable and tissue adhesive EGCG-laden hydrogel depot (EGCG HYPOT), which can achieve anti-inflammatory and antioxidative effects via smart delivery of EGCG. Specifically, the phenylborate ester bonds, formed by EGCG and PBA-modified methacrylated hyaluronic acid (HAMA-PBA), endow EGCG HYPOT injectability, shape adaptation and efficient load of EGCG. After photo-crosslinking, EGCG HYPOT exhibits good mechanical properties, tissue adhesion and sustained acid-responsive release of EGCG. EGCG HYPOT can scavenge oxygen and nitrogen free radicals. Meanwhile, EGCG HYPOT can scavenge intracellular reactive oxygen species (ROS) and suppress the expression of pro-inflammatory factors. EGCG HYPOT may provide a new idea for alleviation of inflammatory disturbance.

1. Introduction

Oxidative stress and inflammation are common pathological mechanisms for the onset and progression of tissue degeneration, such as intervertebral disc degeneration (IVDD), osteoarthritis and periodontitis (Adams & Roughley, 2006; Cao et al., 2022; Lyu et al., 2021; Schepper et al., 2010). Thereinto, IVDD is a common clinical disease worldwide and about 60 % to 80 % of population suffer from the disease during lifetime (Chen et al., 2022; Fujii et al., 2019; Vo et al., 2016). It may be an urgent and promising therapy for IVDD by regulating the oxidative stress and inflammation microenvironment in IVD.

Extensive researches have been focusing on the efficient loading and controlled release of active ingredients to ameliorate inflammation (Amani et al., 2019; Hartvigsen et al., 2018; Kamali et al., 2021). As a catechin-like polyphenol monomer isolated from tea leaves, EGCG features antioxidant, anti-inflammatory, antibacterial, anti-angiogenic, anti-enzyme degradation and anticancer properties, which has been applied in wound healing, cardiovascular disease, neurodegenerative diseases and osteoarthritis fields (Abdolahad et al., 2013; Ahmed, 2010; Akhavan et al., 2012; Chu et al., 2017; Khalatbary & Khademi, 2020; Xu et al., 2021). It has been reported that EGCG can scavenge ROS and inhibit the expression of pro-inflammatory mediators (Han et al., 2012; Krupkova et al., 2014; Shi et al., 2017; Song et al., 2021). Thus, EGCG is a competent natural small molecule drug for the treatment of IVDD. As the avascular nature of IVD, oral administration is difficult to exert the therapeutic effect of EGCG (Othman et al., 2017). Local drug delivery is an alternative to improve drug utilization (El-Mowafy et al., 2010). However, direct injection of EGCG has many problems, such as easy inactivation by oxidation, toxicity from high local concentration and off-target to surrounding tissues, leading to failure of long-lasting and

* Corresponding authors at: Department of Biomedical Engineering, School of Materials Science and Engineering, South China University of Technology, Guangzhou 510006, PR China.
E-mail addresses: feng2119@scut.edu.cn (Q. Feng), caoxd@scut.edu.cn (X. Cao).
[1] Contribute equally to this work.

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effective treatment (Krupkova et al., 2016; Mavlyanova et al., 2020; Tryfonidou et al., 2020). Therefore, it is the prerequisite for IVDD therapy to explore the suitable EGCG delivery vehicle.

As the main component of IVD, nucleus pulposus (NP) is a gel-like structure (Priyadarshani et al., 2016). From the perspective of bionics, hydrogel is considered to be an ideal material for the treatment of IVDD due to its viscoelasticity similar to natural NP. In order to reduce the invasion of surgery and adapt to irregular NP defects, injectable hydrogel has become a research hotspot in the repair of IVDD (Culbert et al., 2022). To avoid cytotoxicity of degradation products and complex synthesis process, injectable hydrogel based on phenylboronic acid-diol dynamic covalent interaction has attracted much attention in recent years (De Oliveira et al., 2020; He et al., 2011; Huang et al., 2018; Yesilyurt et al., 2016; Zhao et al., 2021). In theory, due to the capability of forming borate esters with 1,2- and 1,3-diol, phenylboronic acid groups can bond with EGCG and protect its activity (Huang et al., 2018; Narkar et al., 2016). Therefore, EGCG delivery system based on borate ester complexation can provide an efficiently loading and protective mechanism for EGCG to prevent irreversible oxidative inactivation and improve bioavailability. The catechin-like properties of EGCG endows hydrogels with good tissue adhesion so that EGCG can be stably retained at the injection site (Kim et al., 2020; Zhao et al., 2021). In addition, the phenylboronic acid-diol complex has pH-responsive property and dissociates at acidic pH, which can ensure smart EGCG delivery at the acidic environment of IVDD (Guo et al., 2015; He et al., 2011). However, the stability of injectable hydrogels is usually poor, which limits their application in IVDD (Guvendiren et al., 2012; Uman et al., 2020). Double crosslinking is a significant method to improve the stability of hydrogel and cure IVDD. Hence, injectable and tissue adhesive hydrogel with satisfactory stability and smart EGCG delivery, is theoretically a viable tool to control oxidative stress and inflammation, eventually achieve IVDD treatment.

Hyaluronic acid (HA) is one of the main components of NP extracellular matrix, which has excellent water retention ability and stimulates the synthesis of extracellular matrix (Kim et al., 2015). In order to improve the cell adhesion of HA, some studies have coupled it with gelatin (Gel) (Shu et al., 2006). Gel is the hydrolysis product of collagen, which can induce cell adhesion and proliferation due to its intrinsic integrin binding domain (Eke et al., 2017). In this study, HA and Gel are selected as the main material of EGCG HYPOT to ensure its good biocompatibility and cell affinity. We hypothesize that EGCG HYPOT can be fabricated by phenylborate ester reaction of EGCG and phenylboronic acid, which can achieve anti-inflammatory and antioxidative effects via smart delivery of EGCG. EGCG plays roles as dynamic crosslinking agent and bioactive drugs in the system. Specially, EGCG HYPOT, firstly formed by dynamic crosslinking between EGCG and HAMA-PBA, has good injectability and shape adaptation. After photocrosslinking of HAMA-PBA and methacrylated gelatin (GelMA), EGCG HYPOT can get improved mechanical properties, integrate closely with the tissue, and achieve sustained acid-responsive release of EGCG. In the end, EGCG HYPOT exhibits favorable cytocompatibility, antioxidant and anti-inflammatory effects. This injectable and tissue adhesive EGCG HYPOT brings a promising idea for the treatment of IVDD.

2. Material and methods

2.1. Materials

Sodium hyaluronate (10 MDa) was purchased from Shanghai Crystal Pure industrial Co., Ltd. Gelatin Type A (Gel, porcine skin, ~300 g Bloom) and methacrylic anhydride (MA) were purchased from Sigma-Aldrich. Cell coun ting kit-8 (CCK-8), Calcein/PI Cell Viability/Cytotoxicity Assay Kit and Reactive Oxygen Species Assay Kit were purchased from Beyotime Biotechnology. All other reagents were purchased from Aladdin unless otherwise stated.

2.2. Synthesis and characterization of HAMA-PBA

Briefly, 1 g of sodium hyaluronate was dissolved in 300 mL of deionized water, and 3 mL of methacrylic anhydride (MA) was added under ice bath conditions. After stirring evenly, the pH of the solution was adjusted to 8.0 with 1 M NaOH solution, and then the reaction was continued for 24 h under ice bath conditions. The reaction solution was dialyzed with deionized water (MWCO 8000–14,000 Da) for at least 3 days, and then freeze-dried to obtain HAMA. Then, 1 g of HAMA was dissolved in 300 mL of 2-morpholinoethanesulphonic acid (MES) (0.1 M) buffer solution. 1.4 g of 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride hydrate (DMTMM) was added to activate the carboxyl group on hyaluronic acid and 0.11 g of 3-aminophenylboronic acid was added to react for 24 h. The reaction solution was dialyzed with deionized water (MWCO 8000–14,000 Da) for at least 3 days, and then freeze-dried to obtain HAMA-PBA. The freeze-dried HAMA-PBA was dissolved in D_2O at a concentration of 10 mg/mL. The product structure was determined and the degree of substitution (DS) was calculated by ^1H NMR (Bruker AVANCE 400 MHz). The DS of HAMA-PBA was calculated as equation:

$$DS_{MA}(\%) = \frac{\mathrm{A(MA\ of\ HAMA-PBA)}}{\mathrm{A(methyl\ of\ HA)}} \times \frac{3}{2} \times 100\%$$

$$DS_{PBA}(\%) = \frac{\mathrm{A(PBA\ of\ HAMA-PBA)}}{\mathrm{A(methyl\ of\ HA)}} \times \frac{3}{4} \times 100\%$$

where, A (MA of HAMA-PBA), A (PBA of HAMA-PBA) and A (methyl of HA) are the integrated intensities of vinyl, benzene ring and methyl (-$NHCOCH_3$) in the spectra of HAMA-PBA, respectively.

2.3. Synthesis and characterization of GelMA

Briefly, 10 g of gelatin type A was dissolved in 100 mL of PBS buffer solution, 5 mL of methacrylic anhydride was added dropwise, and the reaction was continued at 50 °C for 3 h. Then the reaction was diluted and terminated by adding PBS buffer solution. The reaction solution was dialyzed in deionized water at 40 °C (MWCO 8000–14,000 Da) for at least 5 days, and then freeze-dried to obtain GelMA. The freeze-dried gelatin and GelMA were dissolved in D_2O at a concentration of 10 mg/mL. The product structure and the DS were determined by ^1H NMR (Bruker AVANCE 400 MHz). The spectra were normalized to the phenylalanine signal (6.9–7.5 ppm). The DS of GelMA was calculated as equation:

$$DS_{MA}(\%) = 1 - \frac{\mathrm{A(lysine\ methylene\ of\ GelMA)}}{\mathrm{A(lysine\ methylene\ of\ gelatin)}} \times 100\%$$

where, A (lysine methylene of gelatin) and A (lysine methylene of GelMA) are the integration area of lysine methylene at 2.8–2.95 ppm in the spectra of unmodified gelatin and GelMA, respectively.

2.4. Preparation of EGCG HYPOT

HAMA-PBA, GelMA and photoinitiator (lithium phenyl(2,4,6-trimethylbenzoyl) phosphinate, LAP) were dissolved in PBS buffer solution, and uniformly mixed with different concentrations of EGCG to obtain single crosslinked EGCG HYPOT (SC EGCG HYPOT), including 1 w/v% HAMA-PBA, 5 w/v% GelMA, 2 w/v‰ LAP, 0.5 or 1 w/v‰ EGCG. The SC EGCG HYPOT was irradiated by UV (5 mW/cm^2, 10 s) to obtain double crosslinked EGCG HYPOT (DC EGCG HYPOT).

2.5. The morphology characterization of EGCG HYPOT

The as-prepared hydrogels were lyophilized overnight and cut off immediately after treating with liquid nitrogen. The morphology of hydrogel network was observed with a scanning electron microscope

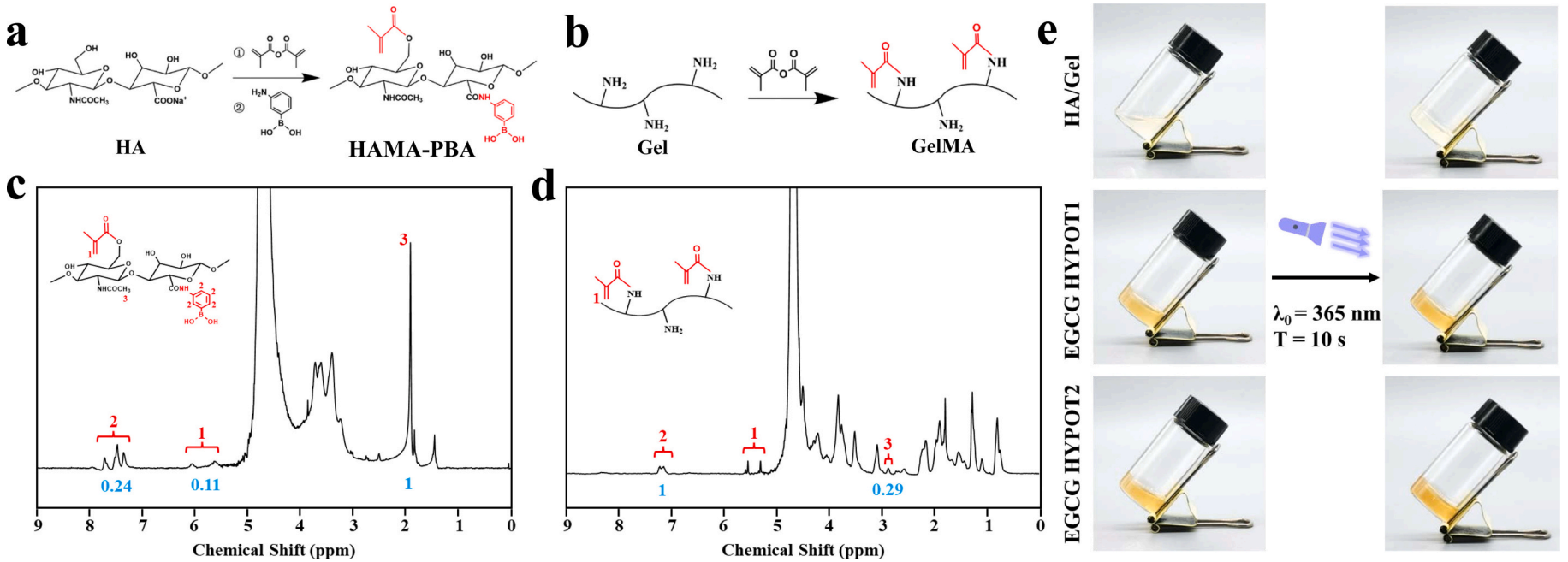


Fig. 1. Design and formation of HAMA-PBA, GelMA and EGCG HYPOT. The scheme of (a) HAMA-PBA and (b) GelMA. The ^{1}H NMR spectra of (c) HAMA-PBA and (d) GelMA, red represents the position of the proton peak, and blue represents the integrated area of the peaks. (e) Diagram of hydrogel gelation.

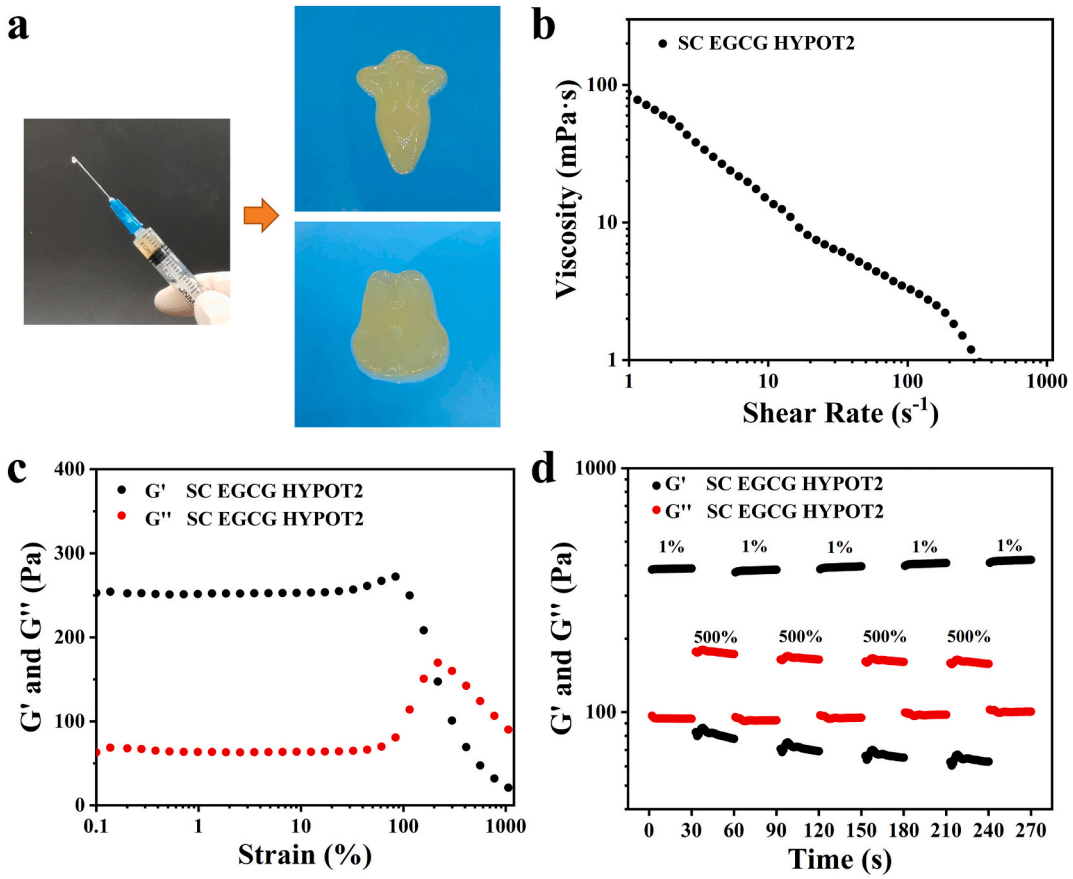


Fig. 2. Injectable, self-healing and shape adaptability of SC EGCG HYPOT2. (a) Macroscopic injectability and shape adaptability. (b) Viscosity, (c) amplitude sweep and (d) cyclic strain sweep measurement of SC EGCG HYPOT2.

(SEM, Quanta 200, Netherlands FEI). The macroscopic injectability, self-healing and shape adaptability of single crosslinked hydrogels were recorded by digital photographs.

2.6. Rheological properties of SC EGCG HYPOT

The rheological properties of SC EGCG HYPOT were characterized by an air oscillatory rheometer (Anton Paar), and 0.5 mL of single crosslinked hydrogels were placed between parallel plates with a diameter of 25 mm. The critical strain point was obtained by strain amplitude scanning test (γ = 0.1 %–1000 %) with fixed angular frequency of 10 rad·s^{-1}. The self-healing performance was characterized by alternating strain scanning test. When the angular frequency was fixed at 10 rad·s^{-1}, the amplitude oscillation strain was switched from low strain (γ = 1 %, 5 intervals of 30 s each) to high strain (γ = 500 %, 4 intervals of 30 s each), and the low strain and high strain alternated for 4 cycles. The shear thinning behavior was proved by viscosity test, and the shear viscosity was measured in the continuous flow with shear rate of 1–1000 s^{-1}.

2.7. Mechanical properties of EGCG HYPOT

The mechanical properties of EGCG HYPOT were analyzed through Dynamic mechanical analyzer (DMA, TA). Specifically, the stress-strain curve was obtained by compressing the cylindrical hydrogel at a strain rate of 100 %/min, and the compression modulus was obtained by calculating the slope of the linear part of the 10 %–15 % strain region. In addition, the mechanical recovery of DC EGCG HYPOT2 was characterized by continuous cyclic compression loading-unloading tests and digital photographs.

2.8. Adhesion properties of EGCG HYPOT

Dynamic mechanical analyzer (DMA, TA) was used for lap shear test to measure the adhesion ability of EGCG HYPOT. After removing subcutaneous fat and hair, fresh pig skins were cut to slices with size of 2 cm × 1 cm. The mechanical strength of SC EGCG HYPOT is weak that it is easily damaged by external force after adhering to the tissue surface. The hydrogel can exhibit stable adhesion properties after photocrosslinking. Therefore, firstly, SC EGCG HYPOT was added between the two pigskins with the bonding area of 1 cm × 1 cm. Then, the samples were in-situ photo-crosslinked, and pulled at a rate of 0.5 N/min until rupture. The bond strength (Pa = N/m^2) was calculated by dividing the maximum force (N) with the bonding area (m^2). The force (F) and displacement (δ) curve was integrated to determine the adhesion energy (W_{adh}), which was normalized by the maximum contact area (A_{max}) with the following equation.

$$W_{adh} = \frac{\int F d\delta}{A_{max}}$$

2.9. Swelling properties of EGCG HYPOT

The cylindrical hydrogels were weighed, immersed in PBS and weighed again every 2 h until the mass no longer increased. The swelling rate of hydrogel S_n (n = 1, 2, 3…) was calculated by the following equation.

$$S_n = \frac{mn - m0}{m0}$$

m_n is the dissolved mass of the hydrogel at different time points, and m_0 is the mass of the hydrogel before soaking in PBS.

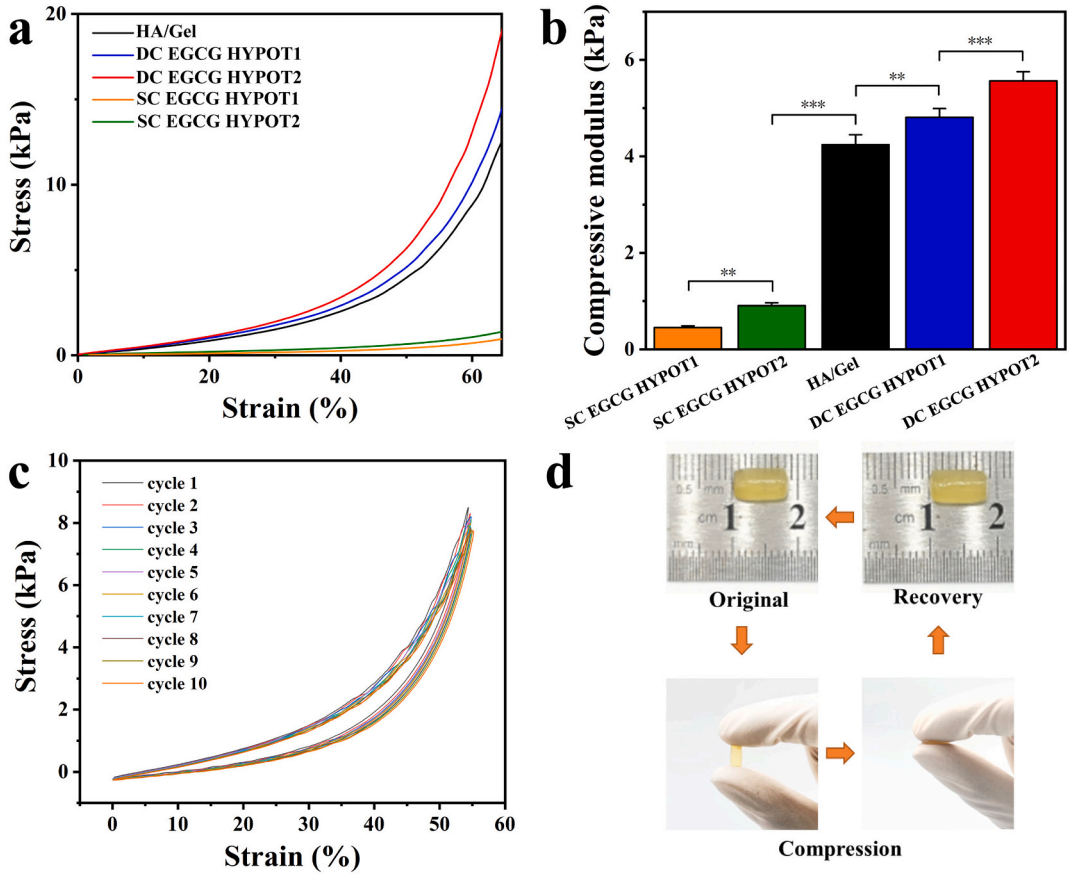


Fig. 3. Mechanical properties of hydrogels. (a) Stress-strain curve; (b) Compression modulus, **P < 0.01, ***P < 0.001, n = 3; (c) Cyclic compression stress-strain curve of DC EGCG HYPOT2; (d) the digital photographs of DC EGCG HYPOT2 under cyclic compression.

2.10. Drug release of EGCG HYPOT

The hydrogels were placed in a shaker (37 °C, 120 rpm) at a concentration of 1 g hydrogel in 20 mL PBS (pH 7.4 or 6.5). The same volume of PBS was added after the supernatant was taken at each time point. The release of EGCG was determined by Folin-Ciocalteu method. Specially, 20 μL of the release solution was completely mixed with 100 μL Folin-Ciocalteu reagent (10 v/v%) in a 96-well plate, followed by the addition of 80 μL of sodium carbonate solution (7.5 w/v%) and left for 60 min at room temperature. The absorbance was measured at 765 nm by a microplate reader (Thermo Scientific).

2.11. Free radical scavenging properties of EGCG HYPOT

DPPH and PTIO scavenging tests were conducted to evaluate the free radical scavenging ability of hydrogels. For DPPH test, the hydrogels were added to DPPH ethanol solution (0.1 mM) at a concentration of 100 mg/mL and incubated for 30 min at room temperature. The absorbance of supernatant was read at 517 nm by a microplate reader (Thermo Scientific). For PTIO test, the hydrogels were added to PTIO aqueous solution (0.6 mM) at a concentration of 100 mg/mL and incubated at 37 °C for 4 h. The absorbance of the supernatant was read at 557 nm by a microplate reader (Thermo Scientific).

2.12. Cytocompatibility of EGCG HYPOT

The cytocompatibility of the hydrogels was assessed by live/dead staining test and cell counting kit-8 (CCK-8). The hydrogels were in situ crosslinked at the bottom of 48-well plate and human nucleus pulposus cells (hNPCs, iCell Bioscience Inc) were co-cultured with the hydrogels at a density of 4×10^4 cells/well. For the live/dead staining test, Calcein and PI in the live/dead staining kit were diluted 1000 times with PBS buffer solution (Gibco) after 1 day of culture, added into the well plate and incubated at 37 °C for 30 min. Then, the cell viability was observed by confocal laser scanning microscopy (CLSM, Leica). For the CCK-8 assay, after 1, 3 and 7 days of culture, CCK-8 diluted solution was added and incubated for 1 h at 37 °C, followed by measuring the absorbance at 450 nm by a microplate reader (Thermo Scientific).

2.13. Antioxidant properties of EGCG HYPOT

The intracellular ROS scavenging ability and cytoprotective effect of hydrogels were investigated by H_2O_2 induced oxidative stress. For the tests of intracellular ROS scavenging ability, the hydrogels were co-cultured with hNPCs in 24-well plate with a density of 8×10^4 cells/well. After being treated with H_2O_2 for 3 h, the cells were gently washed three times with PBS buffer. Then DCFH-DA diluted solution (10 μM) was added and incubated at 37 °C for 20 min, followed by observing intracellular ROS levels at inverted fluorescence microscope (ZEISS). For the tests of cytoprotection, hydrogels were co-cultured with hNPCs in 24-well plate with a density of 4×10^4 cells/well. After being treated with H_2O_2 for 12 h, the cells were gently washed three times with PBS buffer. Calcein and PI were diluted 1000 times with PBS buffer solution, added into the plate and incubated at 37 °C for 30 min. The cytoprotection of hydrogels on hNPCs under oxidative stress environment was observed by inverted fluorescence microscope (ZEISS).

2.14. Anti-inflammatory properties of EGCG HYPOT

The hydrogels were co-cultured with hNPCs in 24-well plate with a

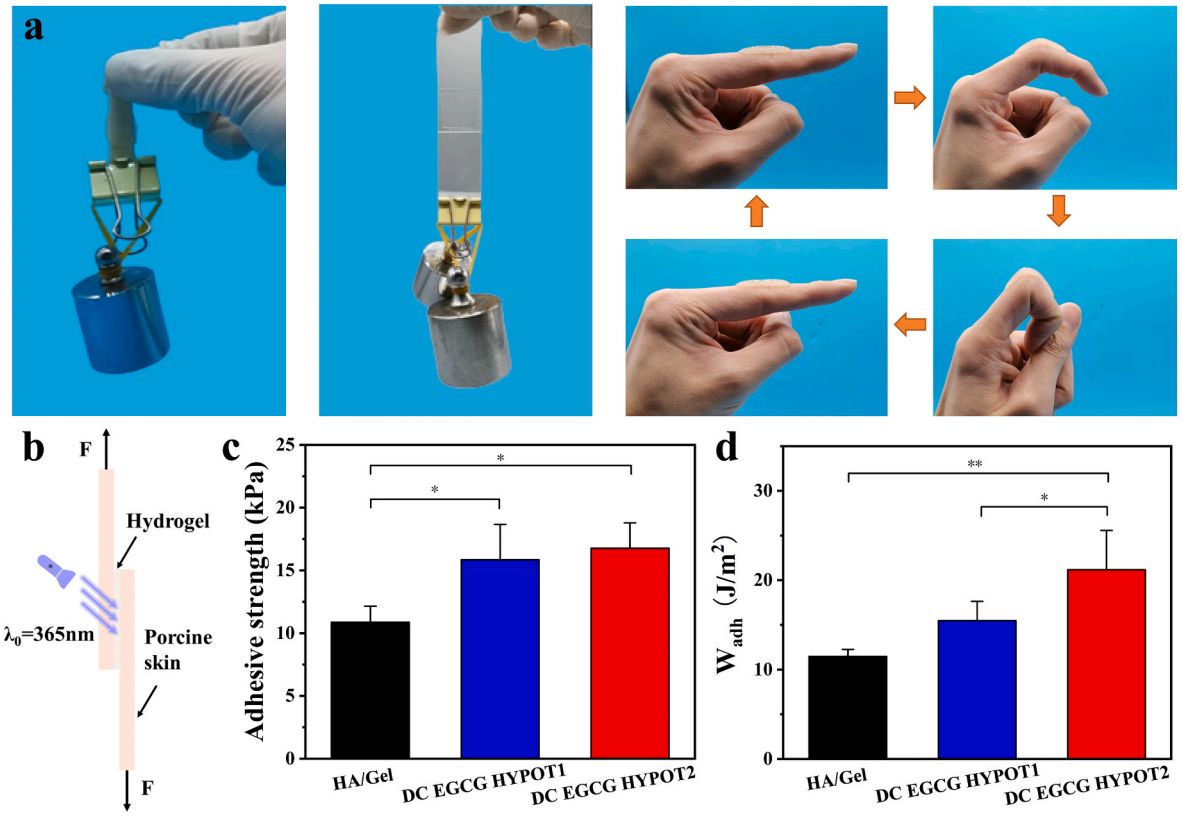


Fig. 4. Adhesion properties of hydrogel. (a) Adhesion of EGCG HYPOT2 to different substrates; (b) schematic diagram of pig skin lap shear test; (c) adhesion strength of pig skin lap shear test, *P < 0.05, n = 3; (d) adhesion work of pig skin lap shear test, *P < 0.05, **P < 0.01, n = 4.

density of 8×10^4 cells/well. IL-1β was added to induce inflammatory response, and the anti-inflammatory effect of the hydrogels was investigated by detecting the expression of inflammation-related genes by PCR assay after 3 days. The target genes included matrix metalloproteinase 3 (MMP-3), matrix metalloproteinase 13 (MMP-13), interleukin 6 (IL-6) and tumor necrosis factor α (TNF-α). Firstly, RNA was extracted using HiPure Total RNA Micro Kit (Magen). Then, the concentration and purity of RNA were determined by NanoDrop 2000. Subsequently, RNA was reverse transcribed into complementary DNA (cDNA) using the PrimeScript RT reagent Kit (Takara). Finally, RT-qPCR was performed using the SYBR green system (Genecopoeia) under the following test conditions: denaturation at 95 °C for 2 min, followed by 40 cycles of annealing at 40 °C for 15 s and extension at 60 °C for 60 s. The relative quantification of target genes was normalized by GAPDH and calculated using the $2^{-\Delta\Delta Ct}$ method. The primer sequences of the target genes are shown in Table S1.

2.15. Statistical analysis

The results of quantitative data were expressed as mean ± standard deviation, and the results were statistically analyzed by one-way analysis of variance (ANOVA) followed by Tukey's post hoc test (SPSS software). Differences were considered statistically significant at $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.001$ (***).

3. Results and discussion

3.1. Design and formation of EGCG HYPOT

As shown in Fig. 1a, HAMA-PBA was synthesized in two steps. Firstly, methacrylic acid was grafted to hyaluronic acid by esterification reaction to obtain HAMA, and then 3-aminophenylboronic acid was grafted to hyaluronic acid by amidation reaction to obtain HAMA-PBA. ^{1}H NMR spectrum in Fig. 1c demonstrates the successful dual modifications of HA. The proton signals at 5.62 and 6.05 ppm are the vinyl characteristic peaks of methacrylic acid. The proton signals at 7.34, 7.47, 7.52 and 7.71 ppm are the benzene ring characteristic peaks of phenylboronic acid. The DS of methacrylic acid and phenylboronic acid are 16.5 % and 18 % respectively by comparing with the integrated peak area of methylene protons peak of HAMA-PBA at 1.9 ppm. Similarly, GelMA was obtained through the reaction between methacrylic anhydride and gelatin (Fig. 1b). Fig. 1d shows the ^{1}H NMR spectra of GelMA. The proton signals at 5.32 and 5.55 ppm are the vinyl characteristic peaks of methacrylic acid, indicating that GelMA is successfully synthesized. The DS of methacrylic acid is 60 % by comparing the lysine methylene characteristic peaks of GelMA with that of gelatin (Fig. S1).

Theoretically, one molecule of EGCG can combine with two molecules of PBA to form a phenylboronic ester bond (Huang et al., 2018; Zhao et al., 2021). We designed the molar ratios of PBA to EGCG as 4:1 and 2:1 to obtain single crosslinked EGCG HYPOT (SC EGCG HYPOT), named SC EGCG HYPOT1 and SC EGCG HYPOT2, to investigate the effects of different EGCG concentrations on the hydrogels. The blank hydrogel (HA/Gel) is free of EGCG. As shown in Fig. 1e, HA/Gel is liquid at physiological pH before photo-crosslinking, while the SC EGCG HYPOT is in the gel state due to the formation of phenylboronic acid ester bond between EGCG and phenylboronic acid group. After 10 s of UV irradiation, HA/Gel turned from liquid to gel state, indicating the photo-crosslinking happened among the double bonds on HAMA-PBA and GelMA. Meanwhile, SC EGCG HYPOT changes into double crosslinked EGCG HYPOT (DC EGCG HYPOT). As shown in Figs. S2 and S3, after photocrosslinking, the vinyl groups in the hydrogels are disappeared and transformed into C—C bonds, and the crosslinking density increase significantly.

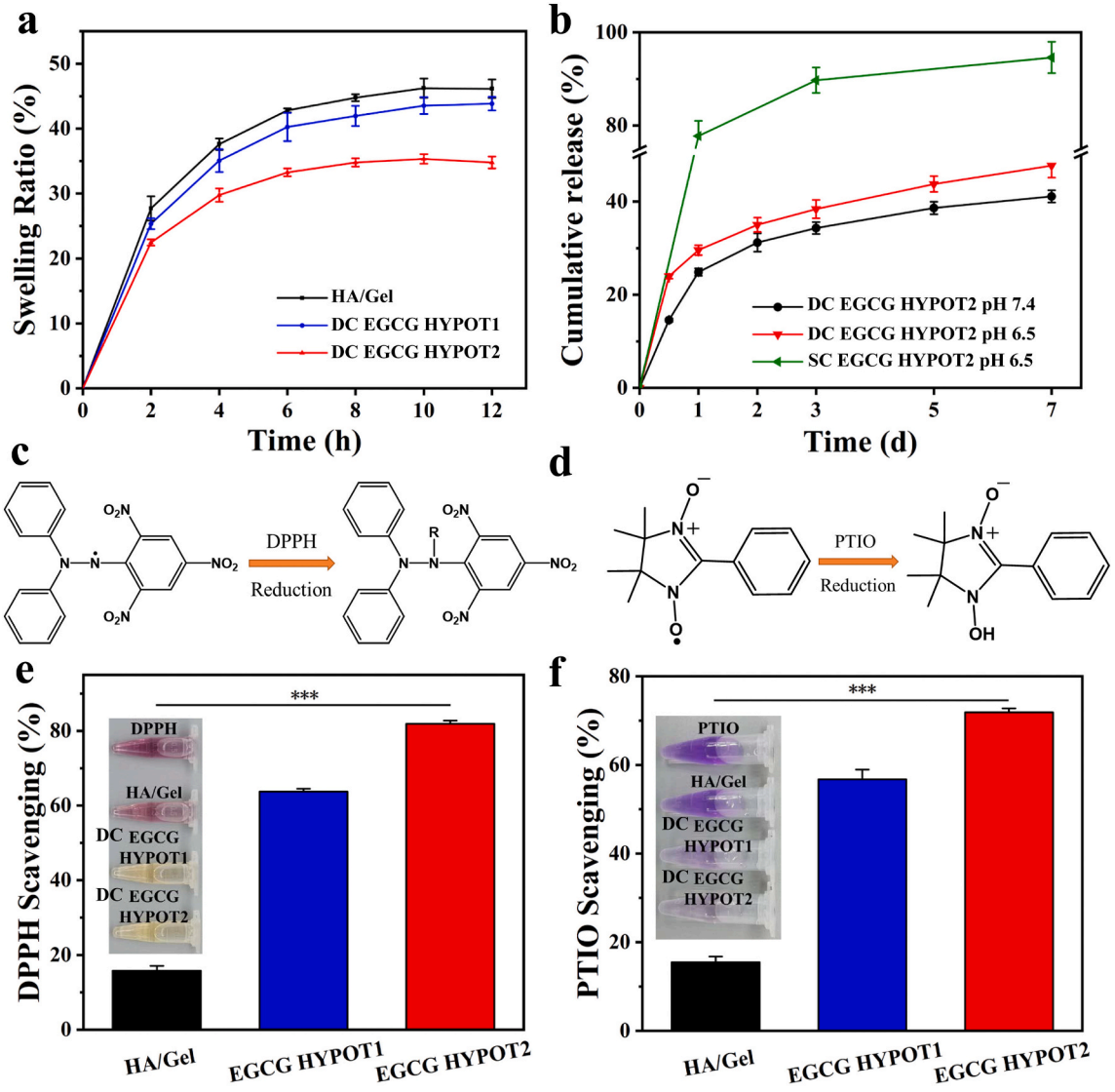


Fig. 5. Drug release and free radical scavenging of hydrogels. (a) Swelling ratio of hydrogel. (b) Release properties of EGCG HYPOT2. Scavenging mechanism of (c) DPPH and (d) PTIO free radical. Scavenging ratio of hydrogel on (e) DPPH and (f) PTIO free radical, ***P < 0.001, n = 4.

3.2. *Injectability, self-healing and shape adaptation of EGCG HYPOT*

Injectability is vital for the invasive treatment of IVDD. A 20 G needle (0.91 mm diameter) is reported to be the largest needle that can be used without risk of damaging to annulus fibrosus (Culbert et al., 2022). As shown in Fig. 2a, SC EGCG HYPOT can be easily injected through a 23 G needle without clogging and quickly heal, which can adapt to irregular tissue defects.

Rheological characterization was used to evaluate the shear thinning and self-healing behavior of SC EGCG HYPOT. To evaluate the injectability of SC EGCG HYPOT, the effect of shear rate on gel viscosity was explored. As shown in Figs. 2b and S4a, the viscosity of SC EGCG HYPOT decreases sharply with increasing shear rate, indicating the typical shear thinning behavior and injectability of SC EGCG HYPOT. The strain scan tests (Figs. 2c and S4b) show that the G′ (storage modulus) and G″ (loss modulus) of SC EGCG HYPOT remain almost constant until the strain reaches 60 %, indicating that SC EGCG HYPOT can maintain a relatively large elastic deformation. With the increase of strain, the hydrogel network becomes unstable. For SC EGCG HYPOT2, G″ is larger than G′ when the strain is over 220 %, indicating that the structure of SC EGCG HYPOT2 collapses and changes from a gel state to a fluid state. As for SC EGCG HYPOT1, hydrogel structure collapses at 400 % of strain. After determining the minimum destructive strain of SC EGCG HYPOT, cyclic strain scans were performed to characterize the self-healing behavior of the hydrogels (Figs. 2d and S4c). When a destructive shear strain (500 %, 30 s) is applied, the G′ value decreases and is lower than the G″ value, indicating that hydrogel network is damaged. When switching to low strain (1 %, 30 s), G′ and G″ values recover, indicating that the hydrogel network is reconstructed. Within a few seconds after multiple shear cycles, SC EGCG HYPOT2 still exhibits 100 % self-healing efficiency, while SC EGCG HYPOT1 hydrogel exhibits 90 % self-healing efficiency. The excellent injectability and self-healing ability of SC EGCG HYPOT is mainly due to the dynamic association and dissociation of phenylboronic acid ester bonds. SC EGCG HYPOT2 can form more dynamic covalent bonds, which make it recover the original state more quickly after being destroyed. The fast self-healing properties ensure EGCG HYPOT to avoid the risk of leakage after injection into NP and match different shapes of defects (Gan et al., 2017; Yu et al., 2020; Zeng et al., 2015). The subsequent photo-initiation radical polymerization can rapidly occur at any desired time. After photo-initiation radical polymerization, DC EGCG HYPOT get improved stability and mechanical strength, but lost injectability. Therefore, the rheological properties of DC EGCG HYPOT weren't investigated.

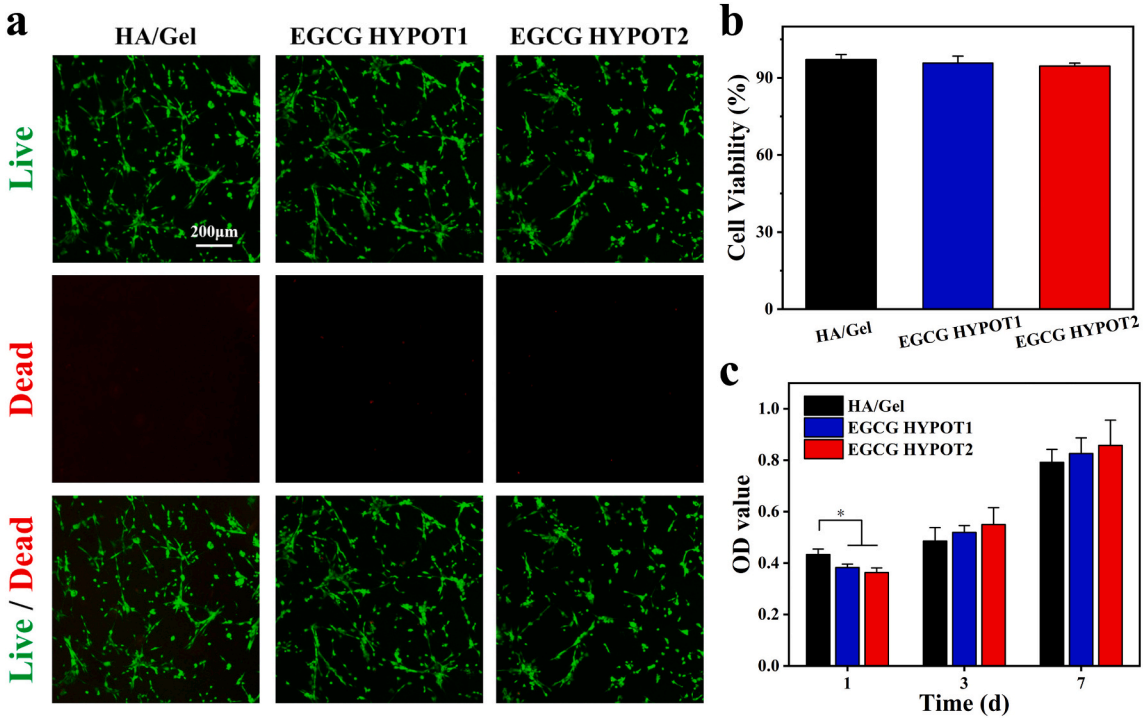


Fig. 6. Biocompatibility of EGCG HYPOT. (a) hNPCs proliferation on all the scaffolds according to the CCK-8 assay results. (b) CLSM images of hNPCs on the hydrogel scaffolds after 1 d of culture. Scale bars: 200 μm. (c) Cell viability, *P < 0.05, n = 3.

3.3. Mechanical properties of EGCG HYPOT

Normally, IVD is in a state of different stress stimulation all the time. The main function of the NP is to absorb and distribute pressure, it is well accepted that hydrogels should have sufficient mechanical strength to resist the continuous compressive load (Bowles & Setton, 2017; Silva-Correia et al., 2013). To characterize the mechanical properties of hydrogels, especially DC EGCG HYPOT, the compressive stress-strain tests and cyclic compression loading-unloading tests were carried out. The stress-strain curve (Fig. 3a) shows that all hydrogels can withstand 65 % of the strain without rupture, indicating their good compressive resistance and high ductility. Fig. 3b shows that the compressive modulus of SC EGCG HYPOT1, SC EGCG HYPOT2, HA/Gel, DC EGCG HYPOT1 and DC EGCG HYPOT2 are 0.45, 0.91, 4.24, 4.81 and 5.57 kPa, respectively. The results show that the mechanical properties of EGCG HYPOT are significantly improved by photo-crosslinking and the increase of EGCG content. Moreover, the compressive modulus of DC EGCG HYPOT2 is similar to the reported modulus of the human NP (Cloyd et al., 2017). The mechanical recovery of hydrogels was characterized by 10 consecutive cycles of compression loading-unloading tests. As shown in Fig. 3c, the cyclic stress-strain curve of DC EGCG HYPOT2 has a high degree of overlap, indicating the hydrogel own good mechanical recovery properties. During the loading-unloading cycle, the compression stress-strain curve exhibited some hysteresis, and the maximum compressive stress decreasing by 9 % after 10 cycles. This is mainly attributed to the energy dissipation of phenylboronic acid ester bonds (Yuan et al., 2021). The deformation recovery performance is also confirmed by the digital photographs of DC EGCG HYPOT2 under repeated compression (Fig. 3d). In a word, the above results indicate that the EGCG HYPOT has good mechanical properties, which enable it to maintain structure stability in IVD.

3.4. Adhesion properties of EGCG HYPOT

To avoid the leakage of implant, it is necessary to investigate the bioadhesive properties of hydrogels (Wiltsey et al., 2013; Zhang et al., 2020; Zhang et al., 2021). As shown in Fig. 4a, DC EGCG HYPOT demonstrates good adhesion to porcine skin, glass and human skin tissues. DC EGCG HYPOT can pick up a 200 g weight when placed between two pieces of pigskins and a 700 g weight when placed between two slides. In addition, DC EGCG HYPOT can firmly adhere on finger joints even after repeated bending.

In order to quantify the adhesion properties of DC EGCG HYPOT, lap shear tensile tests were performed (Fig. 4b). As shown in Fig. 4c, the adhesion strength of HA/Gel, DC EGCG HYPOT1 and DC EGCG HYPOT2 to pigskins are 10.86, 15.84 and 16.77 kPa, respectively. As shown in Figs. 4d and S5, the adhesion energy was calculated from the force-displacement curves as 11.47, 15.47 and 21.18 J/m^2, respectively. The adhesion of HA/Gel hydrogel is based on its hydrophilicity, physical entanglement and electrostatic interactions. During the process of tissue adhesion, the hydrophilic HA/Gel hydrogel firstly absorb and eliminate water on the tissue surface, which is conducive to the full contact of HA/Gel hydrogel with the tissue (Cui et al., 2020). Then, HA/Gel hydrogel binds to tissues through non-covalent interactions such as physical entanglement and electrostatic interactions (Nam & Mooney, 2021). Besides physical entanglement and electrostatic interaction, EGCG can form covalent bonds and hydrogen bonds with different nucleophiles on the tissue surface, which further strengthen the adhesive ability of EGCG HYPOT (Wang et al., 2020). In summary, EGCG HYPOT has good adhesion ability, and the adhesion ability is enhanced as the amount of EGCG increases. Good adhesion ability helps EGCG HYPOT to integrate with the tissue in situ after injection into the target site, avoiding the risk of leakage.

3.5. Drug release and free radical scavenging of EGCG HYPOT

The delivery and antioxidant efficiency of EGCG HYPOT are in correlation with its swelling and release performance. The swelling properties of hydrogels were explored by weighing. As shown in Fig. 5a, HA/Gel, DC EGCG HYPOT1 and DC EGCG HYPOT2 achieved swelling equilibrium in PBS after 12 h, and the equilibrium swelling ratios are 46 %, 43 % and 35 %, respectively. The experimental results show that the

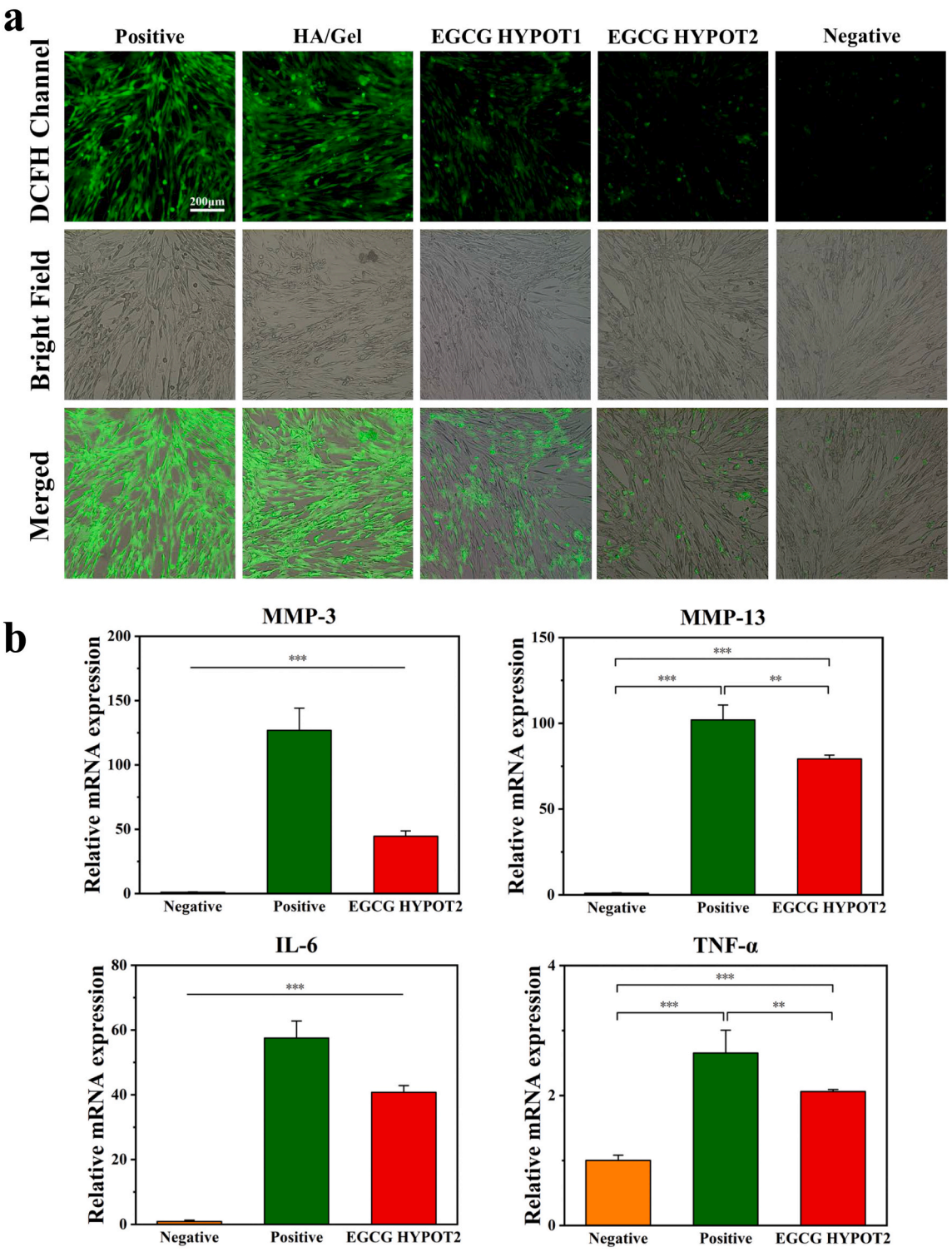


Fig. 7. Antioxidant and anti-inflammatory properties of EGCG HYPOT. (a) DCFH-DA staining assay to characterize the intracellular ROS scavenging ability of hydrogels. Scale bars: 200 μm. (b) Real-time PCR analysis of MMP-3, MMP-13, IL-6 and TNF-α genes (3 d), $^{**}P < 0.01$, $^{***}P < 0.001$, n = 4.

equilibrium swelling ratio decreases with the increase of EGCG content, as phenylboronic acid ester bonds lead to the increase of crosslinking density. Excessive swelling of hydrogels in the target site may increase local pressure and cause tissue defects. Therefore, maintaining a low swelling ratio can make hydrogels more stably to play a mechanical supporting role.

The release of EGCG was performed in PBS and characterized by the Folin-Ciocalteu method. As shown in Figs. 5b and S6, SC EGCG HYPOT showed obvious burst release, with 86 % release of SC EGCG HYPOT1 and 78 % release of SC EGCG HYPOT2 within 1 day. Due to the dynamic properties of phenylboronic acid ester bond, SC EGCG HYPOT does not have swelling equilibrium, but swells until degradation, which results in the obvious burst release behavior. The burst release of EGCG leads to high local concentration, which may cause cytotoxicity and interfere for subsequent treatment. After photo-crosslinking, the burst release of EGCG is significantly reduced. Moreover, DC EGCG HYPOT has faster

release rate and higher cumulative release amount of EGCG at acidic pH 6.5 compared with that in pH 7.4. The dynamic phenylboronic acid ester bonds are pH-responsive and tend to dissociate at acidic pH, so EGCG is released. The pH responsiveness is matched to the acidic microenvironment of IVDD. EGCG HYPOT could be a good carrier of EGCG for the treatment of IVDD.

Oxidative stress plays a key role in aggravating IVDD by strengthening inflammatory responses, disrupting cellular function and accelerating extracellular matrix degradation (Tian et al., 2020). Under normal physiological conditions, oxidants and antioxidants in the body maintain a redox balance. When the intracellular balance of pro-oxidants and antioxidants is disturbed and tilted toward pro-oxidants, it leads to oxidative stress (Hartvigsen et al., 2018). There are two main forms of free radicals in vivo, e.g., reactive oxygen species (ROS) and reactive nitrogen species (RNS) (Hartvigsen et al., 2018). DPPH is a typical nitrogen radical and PTIO is a typical oxygen radical. These two free radical models were used to assess the RNS and ROS scavenging ability of EGCG HYPOT (Fig. 5c and d). As shown in Fig. 5e and f, the DPPH scavenging efficiencies of HA/Gel, DC EGCG HYPOT1 and DC EGCG HYPOT2 are 15.8 %, 63.7 % and 81.9 %, respectively. Meanwhile, those on PTIO are 15.4 %, 56.4 % and 71.8 %, respectively. The digital photographs show the same trend. These results indicate that EGCG HYPOT can effectively scavenge free radicals and resist the effect of oxidative stress on IVDD.

3.6. Biocompatibility of EGCG HYPOT

To evaluate the biocompatibility of EGCG HYPOT and its potential application in the treatment of IVDD, hNPCs were seeded on the hydrogel, followed by assessing the viability and proliferation of hNPCs. It should be noted that EGCG HYPOT refers to DC EGCG HYPOT if it's not specified in the later sections. Cell viability and cell morphology of hNPCs cultured on hydrogels for 1 day were characterized by live/dead staining. Green fluorescence represents live cells and red fluorescence represents dead cells. As shown in Fig. 6a, hNPCs on hydrogels show good viability and similar adhesion level. Cells tend to aggregate into clusters and show a spreading state. Meanwhile, the results of live/dead staining were quantified by Image J software (Fig. 6b). Cell viability of all groups are over 90 %, and there was no significant difference. As shown in Fig. 6c, hNPCs presents normal proliferation on all groups. The cell proliferation ability of the EGCG HYPOT group is little lower than that of the HA/Gel hydrogel group on 1 day. With increase of culture time, there is no significant difference between the OD values of all groups. The above results indicate that EGCG HYPOT have good cytocompatibility and can promote the spreading of hNPCs, which means it can furtherly be used as scaffold for IVDD.

3.7. Antioxidant and anti-inflammatory properties of EGCG HYPOT

Oxidative stress caused by excessive ROS plays an important role in IVDD. Under oxidative stress, DNA, lipids, proteins and other cellular components are destroyed, thereby increasing cell apoptosis (Feng et al., 2017). In order to evaluate the ROS scavenging ability of EGCG HYPOT, the intracellular ROS level was detected by fluorescent probe DCFH-DA. H_2O_2 is a common agent used to induce oxidative stress, which can induce ROS production. As shown in Fig. 7a, the green fluorescence intensity of the positive control group treated with H_2O_2 is significantly higher than that of the negative control group, indicating that H_2O_2 can successfully induce excess ROS production in hNPCs. The green fluorescence intensity of EGCG HYPOT group decreases significantly, proving that EGCG HYPOT can effectively inhibit ROS production. Especially, the fluorescence intensity of EGCG HYPOT2 group is visually close to that of negative control group, proving that it has the strongest intracellular ROS scavenging ability. Meanwhile, the results of DCFH-DA staining were quantified by Image J software (Fig. S7). The fluorescence intensity decrease significantly with the increase of EGCG content. In addition, we also verified the protective effect of EGCG HYPOT on hNPCs under oxidative stress environment by live/dead staining. As shown in Fig. S8, the number of live cells significantly reduces in the positive control group, which representing the apoptosis of hNPCs induced by H_2O_2. Similar result is found in the HA/Gel group. However, for the EGCG HYPOT groups, the viability of hNPCs is significantly enhanced (Fig. S8). EGCG HYPOT2 shows the best protective effect against oxidative damage. It is reasonable to infer that EGCG HYPOT is qualified to protect hNPCs from the oxidative stress. Combined with the results of DCFH-DA staining and live/dead staining, the excellent antioxidation behavior is ascribed to the dynamic release of EGCG and this performance is concentration-dependent.

Inflammation has been extensively studied as a common pathological mechanism for the progression of IVDD, and interleukin 1β (IL-1β) is the most fully studied pro-inflammatory cytokine associated with IVDD (Le Maitre et al., 2005). The inflammation of hNPCs were induced by IL-1β (Sakai & Grad, 2015). The anti-inflammatory effect of EGCG HYPOT2 was explored through using hNPCs treated with IL-1β as positive control and hNPCs without any treatment as negative control. The results are shown in Fig. 7b. IL-1β is able to activate inflammation signaling in hNPCs, resulting in significantly enhanced expression of catabolic genes and inflammatory mediator genes, such as MMP-3, MMP-13, IL-6 and TNF-α. In contrast, after being treated with EGCG HYPOT2, hNPCs show significantly down-regulated expression of MMP-3, MMP-13, IL-6 and TNF-α, due to the anti-inflammatory and antioxidative effects of EGCG. It deduces that EGCG HYPOT is competent to inhibit pro-inflammatory signaling and matrix catabolism in hNPCs.

4. Conclusions

In summary, an injectable and tissue adhesive EGCG HYPOT is successfully fabricated, which can achieve anti-inflammatory and antioxidative effects. The SC EGCG HYPOT has good injectability, self-healing and shape adaptability, which is favor for minimally invasive treatment to fill irregular tissue defects. After photo-crosslinking, EGCG HYPOT exhibits good adhesive, mechanical property and pH-responsive, which make it match the IVD mechanical property and accelerate the release of EGCG in acidic IVDD environment. EGCG HYPOT can attenuate the apoptosis and inflammatory factor expression of hNPCs. We believe that EGCG HYPOT can be used as a promising scaffold to treat oxidative stress and inflammation during the IVDD process.

CRediT authorship contribution statement

Zhichao He: Methodology, Formal analysis, Visualization, Investigation, Data curation, Resources, Writing – original draft. **Huitong Luo:** Methodology, Data curation, Writing – original draft. **Zetao Wang:** Data curation, Resources, Investigation. **Dafu Chen:** Supervision, Project administration. **Qi Feng:** Data curation, Writing – review & editing, Supervision. **Xiaodong Cao:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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References

Abdolahad, M., Janmaleki, M., Mohajerzade, S., Akhavan, O., & Abbasi, S. (2013). Polyphenols attached graphene nanosheets for high efficiency NIR mediated photodestruction of cancer cells. *Materials Science and Engineering C, 33*, 1498–1505.

Adams, M. A., & Roughley, P. J. (2006). What is intervertebral disc degeneration, and what causes it? *Spine, 31*, 2151–2161.

Ahmed, S. (2010). Green tea polyphenol epigallocatechin 3-gallate in arthritis: Progress and promise. *Arthritis Research & Therapy, 12*, 208.

Akhavan, O., Kalaee, M., Alavi, Z. S., Ghias, S. M. A., & Esfandiar, A. (2012). Increasing the antioxidant activity of green tea polyphenols in the presence of iron for the reduction of graphene oxide. *Carbon, 50*, 3015–3025.

Amani, H., Habibey, R., Shokri, F., Hajmiresmail, S. J., Akhavan, O., Mashaghi, A., et al. (2019). Selenium nanoparticles for targeted stroke therapy through modulation of inflammatory and metabolic signaling. *Scientific Reports, 9*, 6004.

Bowles, R. D., & Setton, L. A. (2017). Biomaterials for intervertebral disc regeneration and repair. *Biomaterials, 129*, 54–67.

Cao, G., Yang, S., Cao, J., Tan, Z., Wu, L., Dong, F., et al. (2022). The role of oxidative stress in intervertebral disc degeneration. *Oxidative Medicine and Cellular Longevity, 2022*, Article 2166817.

Chen, S., Chen, M., Wu, X., Lin, S., Tao, C., Cao, H., et al. (2022). Global, regional and national burden of low back pain 1990–2019: A systematic analysis of the global burden of disease study 2019. *Journal of Orthopaedic Translation, 32*, 49–58.

Chu, C., Deng, J., Man, Y., & Qu, Y. (2017). Green tea extracts Epigallocatechin-3-gallate for different treatments. *BioMed Research International, 2017*, Article 5615647.

Cloyd, J. M., Malhotra, N. R., Weng, L., Chen, W., Mauck, R. L., & Elliott, D. M. (2017). Material properties in unconfined compression of human nucleus pulposus, injectable hyaluronic acid-based hydrogels and tissue engineering scaffolds. *European Spine Journal, 16*, 1892–1898.

Cui, C., Wu, T., Chen, X., Liu, Y., Li, Y., Xu, Z., et al. (2020). A janus hydrogel wet adhesive for internal tissue repair and anti-postoperative adhesion. *Advanced Functional Materials, 30*, Article 2005689.

Culbert, M. P., Warren, J. P., Dixon, A. R., Fermor, H. L., Beales, P. A., & Wilcox, R. K. (2022). Evaluation of injectable nucleus augmentation materials for the treatment of intervertebral disc degeneration. *Biomaterials Science, 10*, 874–891.

De Oliveira, M. M., Nakamura, C. V., & Auzély-Velty, R. (2020). Boronate-ester crosslinked hyaluronic acid hydrogels for dihydrocaffeic acid delivery and fibroblasts protection against UVB irradiation. *Carbohydrate Polymers, 247*, Article 116845.

Eke, G., Mangir, N., Hasirci, N., MacNeil, S., & Hasirci, V. (2017). Development of a UV crosslinked biodegradable hydrogel containing adipose derived stem cells to promote vascularization for skin wounds and tissue engineering. *Biomaterials, 129*, 188–198.

El-Mowafy, A. M., Al-Gayyar, M. M., Salem, H. A., El-Mesery, M. E., & Darweish, M. M. (2010). Novel chemotherapeutic and renal protective effects for the green tea (EGCG): Role of oxidative stress and inflammatory-cytokine signaling. *Phytomedicine, 17*, 1067–1075.

Feng, C., Yang, M., Lan, M., Liu, C., Zhang, Y., Huang, B., et al. (2017). ROS: Crucial intermediators in the pathogenesis of intervertebral disc degeneration. *Oxidative Medicine and Cellular Longevity, 2017*, Article 5601593.

Fujii, K., Yamazaki, M., Kang, J. D., Risbud, M. V., Cho, S. K., Qureshi, S. A., et al. (2019). Discogenic Back pain: Literature review of definition, diagnosis, and treatment. *JBMR Plus, 3*, Article e10180.

Gan, Y., Li, P., Wang, L., Mo, X., Song, L., Xu, Y., et al. (2017). An interpenetrating network-strengthened and toughened hydrogel that supports cell-based nucleus pulposus regeneration. *Biomaterials, 136*, 12–28.

Guo, J., Sun, H., Alt, K., Tardy, B. L., Richardson, J. J., Suma, T., et al. (2015). Boronate-phenolic network capsules with dual response to acidic pH and cis-diols. *Advanced Healthcare Materials, 4*, 1796–1801.

Guvendiren, M., Lu, H. D., & Burdick, J. A. (2012). Shear-thinning hydrogels for biomedical applications. *Soft Matter, 8*, 260–272.

Han, S. G., Han, S., Tobore, M., & Hennig, B. (2012). EGCG protects endothelial cells against PCB 126-induced inflammation through inhibition of AhR and induction of Nrf2-regulated genes. *Toxicology and Applied Pharmacology, 261*, 181–188.

Hartvigsen, J., Hancock, M. J., Kongsted, A., Louw, Q., Ferreira, M. L., Genevay, S., et al. (2018). What low back pain is and why we need to pay attention. *The Lancet, 391*, 2356–2367.

He, L., Fullenkamp, D. E., Rivera, J. G., & Messersmith, P. B. (2011). pH responsive self-healing hydrogels formed by boronate–catechol complexation. *Chemical Communications, 47*, 7497–7499.

Huang, Z., Delparastan, P., Burch, P., Cheng, J., Cao, Y., & Messersmith, P. B. (2018). Injectable dynamic covalent hydrogels of boronic acid polymers cross-linked by bioactive plant-derived polyphenols. *Biomaterials Science, 6*, 2487–2495.

Kamali, A., Ziadlou, R., Lang, G., Pfannkuche, J., Cui, S., Li, Z., et al. (2021). Small molecule-based treatment approaches for intervertebral disc degeneration: Current options and future directions. *Theranostics, 11*, 27–47.

Khalatbary, A. R., & Khademi, E. (2020). The green tea polyphenolic catechin epigallocatechin gallate and neuroprotection. *Nutritional Neuroscience, 23*, 281–294.

Kim, D. H., Martin, J. T., Elliott, D. M., Smith, L. J., & Mauck, R. L. (2015). Phenotypic stability, matrix elaboration and functional maturation of nucleus pulposus cells encapsulated in photocrosslinkable hyaluronic acid hydrogels. *Acta Biomaterialia, 12*, 21–29.

Kim, S. H., Kim, K., Kim, B. S., An, Y. H., Lee, U. J., Lee, S. H., et al. (2020). Fabrication of polyphenol-incorporated anti-inflammatory hydrogel via high-affinity enzymatic crosslinking for wet tissue adhesion. *Biomaterials, 242*, Article 119905.

Krupkova, O., Ferguson, S. J., & Wuertz-Kozak, K. (2016). Stability of (−)-epigallocatechin gallate and its activity in liquid formulations and delivery systems. *The Journal of Nutritional Biochemistry, 37*, 1–12.

Krupkova, O., Sekiguchi, M., Klasen, J., Hausmann, O., Konno, S., Ferguson, S. J., et al. (2014). Epigallocatechin 3-gallate suppresses interleukin-1β-induced inflammatory responses in intervertebral disc cells in vitro and reduces radiculopathic pain in rats. *European Cells & Materials, 28*, 372–386.

Le Maitre, C. L., Freemont, A. J., & Hoyland, J. A. (2005). The role of interleukin-1 in the pathogenesis of human intervertebral disc degeneration. *Arthritis Research & Therapy, 7*, Article R732.

Lyu, F. J., Cui, H., Pan, H., Mc Cheung, K., Cao, X., Iatridis, J. C., & Zheng, Z. (2021). Painful intervertebral disc degeneration and inflammation: From laboratory evidence to clinical interventions. *Bone Research, 9*, 7.

Mavlyanova, R., Yang, R., Tao, T., Aquib, M., Kesse, S., Maviah, M. B. J., et al. (2020). Injectable hydrogels for targeted delivering of therapeutic molecules for tissue engineering and disease treatment. *Polymers for Advanced Technologies, 31*, 192–203.

Nam, S., & Mooney, D. (2021). Polymeric tissue adhesives. *Chemical Review, 121*, 11336–11384.

Narkar, A. R., Barker, B., Clisch, M., Jiang, J., & Lee, B. P. (2016). pH responsive and oxidation resistant wet adhesive based on reversible catechol-boronate complexation. *Chemistry of Materials, 28*, 5432–5439.

Othman, A. I., El-Sawi, M. R., El-Missiry, M. A., & Abukhalil, M. H. (2017). Epigallocatechin-3-gallate protects against diabetic cardiomyopathy through modulating the cardiometabolic risk factors, oxidative stress, inflammation, cell death and fibrosis in streptozotocin-nicotinamide-induced diabetic rats. *Biomedicine & Pharmacotherapy, 94*, 362–373.

Priyadarshani, P., Li, Y., & Yao, L. (2016). Advances in biological therapy for nucleus pulposus regeneration. *Osteoarthritis and Cartilage, 24*, 206–212.

Sakai, D., & Grad, S. (2015). Advancing the cellular and molecular therapy for intervertebral disc disease. *Advanced Drug Delivery Reviews, 84*, 159–171.

Schepper, E. I. T., Damen, J., van Meurs, J. B. J., Ginai, A. Z., Popham, M., Hofman, A., et al. (2010). The association between lumbar disc degeneration and low Back pain: The influence of age, gender, and individual radiographic features. *Spine, 35*, 531–536.

Shi, J., Deng, H., Pan, H., Xu, Y., & Zhang, M. (2017). Epigallocatechin-3-gallate attenuates microcystin-LR induced oxidative stress and inflammation in human umbilical vein endothelial cells. *Chemosphere, 168*, 25–31.

Shu, X. Z., Ahmad, S., Liu, Y., & Prestwich, G. D. (2006). Synthesis and evaluation of injectable, in situ crosslinkable synthetic extracellular matrices for tissue engineering. *Journal of Biomedical Materials Research Part A, 79*, 902–912.

Silva-Correia, J., Correia, S. I., Oliveira, J. M., & Reis, R. L. (2013). Tissue engineering strategies applied in the regeneration of the human intervertebral disk. *Biotechnology Advances, 31*, 1514–1531.

Song, D., Ge, J., Wang, Y., Yan, Q., Wu, C., Yu, H., et al. (2021). Tea polyphenol attenuates oxidative stress-induced degeneration of intervertebral discs by regulating the Keap1/Nrf2/ARE pathway. *Oxidative Medicine and Cellular Longevity, 2021*, Article 6684147.

Tian, Y., Bao, Z., Ji, Y., Mei, X., & Yang, H. (2020). Epigallocatechin-3-gallate protects H (2)O(2)-induced nucleus pulposus cell apoptosis and inflammation by inhibiting cGAS/Sting/NLRP3 activation. *Drug Design Development and Therapy, 14*, 2113–2122.

Tryfonidou, M. A., de Vries, G., Hennink, W. E., & Creemers, L. B. (2020). "Old drugs, new tricks" – local controlled drug release systems for treatment of degenerative joint disease. *Advanced Drug Delivery Reviews, 160*, 170–185.

Uman, S., Dhand, A., & Burdick, J. A. (2020). Recent advances in shear-thinning and self-healing hydrogels for biomedical applications. *Journal of Applied Polymer Science, 137*, 48668.

Vo, N. V., Hartman, R. A., Patil, P. R., Risbud, M. V., Kletsas, D., Iatridis, J. C., et al. (2016). Molecular mechanisms of biological aging in intervertebral discs. *Journal of Orthopaedic Translation, 34*, 1289–1306.

Wang, L., Zhang, X., Yang, K., Fu, Y. V., Xu, T., Li, S., et al. (2020). A novel double-crosslinking-double-network design for injectable hydrogels with enhanced tissue adhesion and antibacterial capability for wound treatment. *Advanced Functional Materials, 30*, Article 1904156.

Wiltsey, C., Kubinski, P., Christiani, T., Toomer, K., Sheehan, J., Branda, A., et al. (2013). Characterization of injectable hydrogels based on poly(N-isopropylacrylamide)-g-chondroitin sulfate with adhesive properties for nucleus pulposus tissue engineering. *Journal of Materials Science: Materials in Medicine, 24*, 837–847.

Xu, F. W., Lv, Y. L., Zhong, Y. F., Xue, Y. N., Wang, Y., Zhang, L. Y., et al. (2021). Beneficial effects of green tea EGCG on skin wound healing: A comprehensive review. *Molecules, 26*, 6123.

Yesilyurt, V., Webber, M. J., Appel, E. A., Godwin, C., Langer, R., & Anderson, D. G. (2016). Injectable self-healing glucose-responsive hydrogels with pH-regulated mechanical properties. *Advanced Materials, 28*, 86–91.

Yu, C., Gao, H., Li, Q., & Cao, X. (2020). Injectable dual cross-linked adhesive hyaluronic acid multifunctional hydrogel scaffolds for potential applications in cartilage repair. *Polymer Chemistry, 11*, 3169–3178.

Yuan, Y., Shen, S., & Fan, D. (2021). A physicochemical double cross-linked multifunctional hydrogel for dynamic burn wound healing: Shape adaptability, injectable self-healing property and enhanced adhesion. *Biomaterials, 276*, Article 120838.

Zeng, Y., Chen, C., Liu, W., Fu, Q., Han, Z., Li, Y., et al. (2015). Injectable microcryogels reinforced alginate encapsulation of mesenchymal stromal cells for leak-proof delivery and alleviation of canine disc degeneration. *Biomaterials, 59*, 53–65.

Zhang, C., Wu, B., Zhou, Y., Zhou, F., Liu, W., & Wang, Z. (2020). Mussel-inspired hydrogels: From design principles to promising applications. *Chemical Society Reviews, 49*, 3605–3637.

Zhang, X., Li, Z., Yang, P., Duan, G., Liu, X., Gu, Z., et al. (2021). Polyphenol scaffolds in tissue engineering. *Materials Horizons, 8*, 145–167.

Zhao, X., Pei, D., Yang, Y., Xu, K., Yu, J., Zhang, Y., et al. (2021). Green tea derivative driven smart hydrogels with desired functions for chronic diabetic wound treatment. *Advanced Functional Materials, 31*, Article 2009442.